SUPPLEMENTAL MATERIAL

Table S1. Full search phrases used for the respective databases.

Pubmed	486 articles
(subcutaneous ICD OR S-ICD OR transvenous ICD OR TV-ICD OR conventional ICD OR dual-chamber ICD OR single chamber ICD OR endovascular defibrillator) AND (random* OR controlled OR propensity)	
Embase	340 articles
('s-icd':ti,ab,kw OR (('subcutaneous':ti,ab,kw OR 'transvenous':ti,ab,kw) AND ('implantable cardioverter-defibrillator':ti,ab,kw OR 'icd':ti,ab,kw)) OR 'tv-icd':ti,ab,kw OR 'conventional icd':ti,ab,kw OR 'dual-chamber icd':ti,ab,kw OR 'single chamber icd':ti,ab,kw OR 'endovascular defibrillator':ti,ab,kw) AND ('randomized':ti,ab,kw OR 'random':ti,ab,kw OR 'randomised':ti,ab,kw OR 'controlled':ti,ab,kw OR 'propensity':ti,ab,kw)	
Web of Science	476 articles
AB = ((subcutaneous ICD OR S-ICD OR transvenous ICD OR TV-ICD OR conventional ICD OR dual-chamber ICD OR single chamber ICD OR endovascular defibrillator) AND (random* OR controlled OR propensity)) OR TI = ((subcutaneous ICD OR S-ICD OR transvenous ICD OR TV-ICD OR conventional ICD OR dual-chamber ICD OR single chamber ICD OR endovascular defibrillator) AND (random* OR controlled OR propensity)) OR KP = ((subcutaneous ICD OR S-ICD OR transvenous ICD OR TV-ICD OR conventional ICD OR dual-chamber ICD OR single chamber ICD OR endovascular defibrillator) AND (random* OR controlled OR propensity)) OR SU = ((subcutaneous ICD OR S-ICD OR transvenous ICD OR TV-ICD OR conventional ICD OR dual-chamber ICD OR single chamber ICD OR endovascular defibrillator) AND (random* OR controlled OR propensity))	
Cochrane Controlled Register of Trials CENTRAL	34 articles
#1 subcutaneous ICD	
#2 transvenous ICD	
#3 dual-chamber ICD	
#4 single-chamber ICD	
#5 endovascular defibrillator	
#6 #1 #2 #3 OR #4 OR #5	
#7 MeSH descriptor: [Randomized Controlled Trial] explode all trees	
#8 MeSH descriptor: [Propensity Score] explode all trees	
#9 #7 OR #8	
#10 #6 AND 11	

Date searched: September 25, 2021

Table S2. Risk-of-bias analysis of included studies.

Cochrane Risk-of-Bias tool for randomized controlled trials						
Study ID	Randomization process	Deviations from intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported result	Overall
PRAETORIAN (Knops 2020)	+	+	+	+	+	+

Newcastle-Ottawa Scale for nonrandomized trials				
Study ID	Selection	Comparability	Exposure	Total
Brouwer 2016	***	**	**	7
Honarbakhsh 2017	***	**	**	7
POINTED (Palmisano 2021)	***	**	**	7
SIMPLE-EFFORTLESS (Brouwer 2018)	***	**	**	7

+ Denotes low risk of bias.

Legend for Newcastle-Ottawa scale:

- Selection: maximum 4 stars
- Comparability: maximum 2 stars
- Exposure: maximum 3 stars
- Total: maximum 9 stars
- Studies with 7-9 stars are considered to be at low risk of bias.

Figure Legends

95%CI, 95% confidence interval; HR, hazard ratio; MH, Mantel-Haenszel; IV, inverse-variance; S-ICD, subcutaneous implantable cardioverter-defibrillator; TV-ICD, transcutaneous implantable cardioverter-defibrillator

Figure S1. Verification of the proportional-hazards assumption with Schoenfeld residuals.

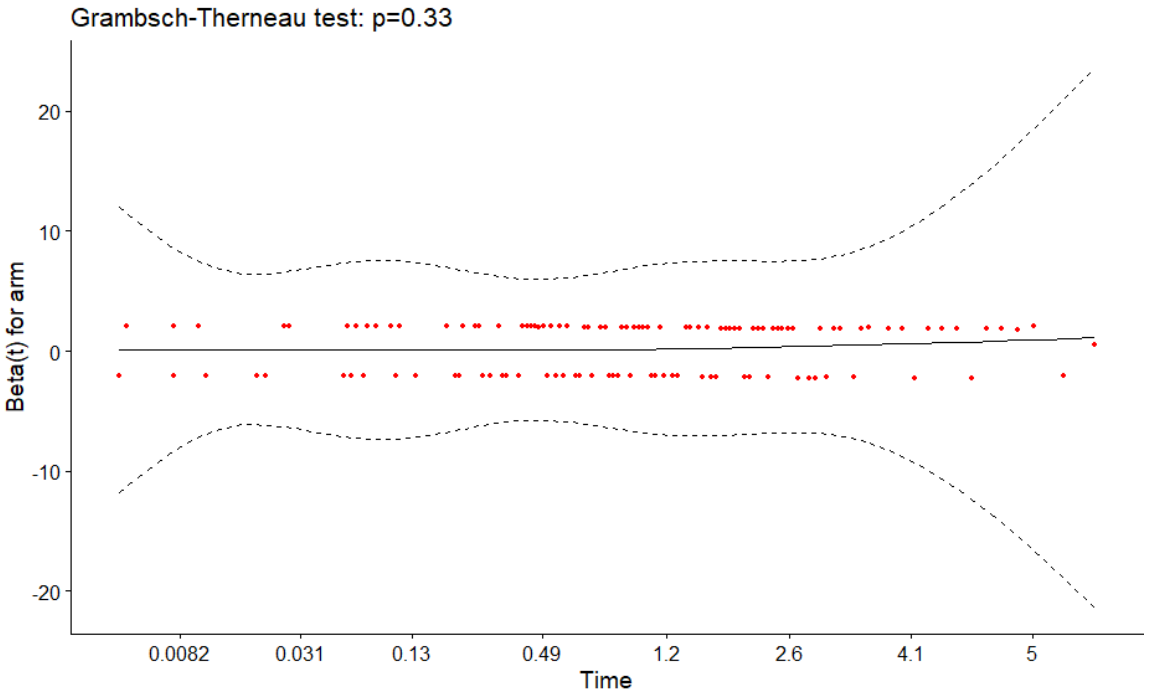


Figure S2. Two-stage Individual Patient Data meta-analysis of device-related complications.

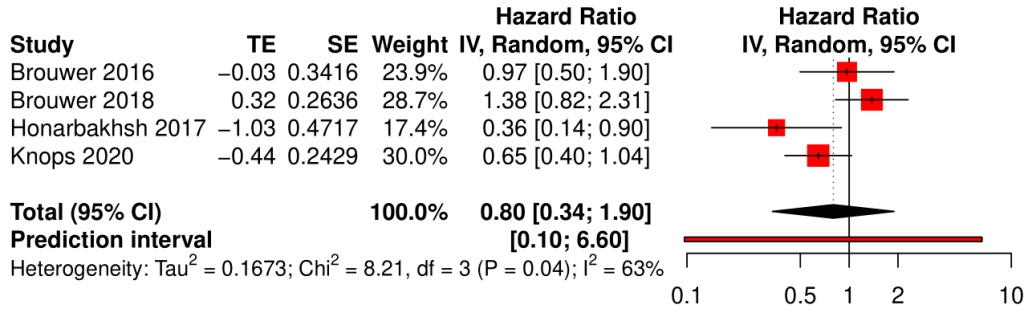


Figure S3. Forest plot of all-cause mortality.

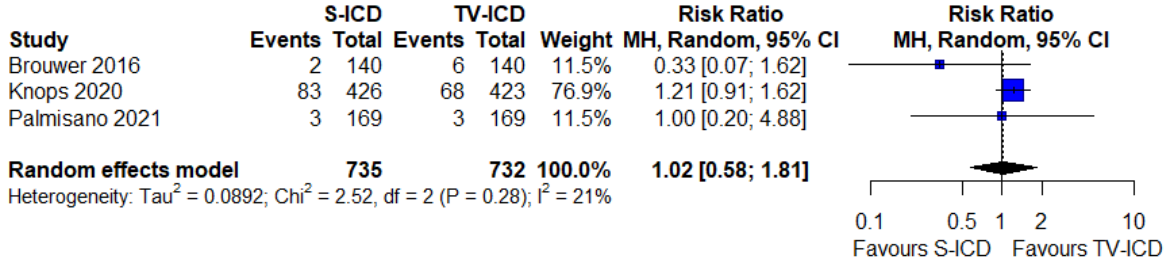


Figure S4. Forest plot of infection.

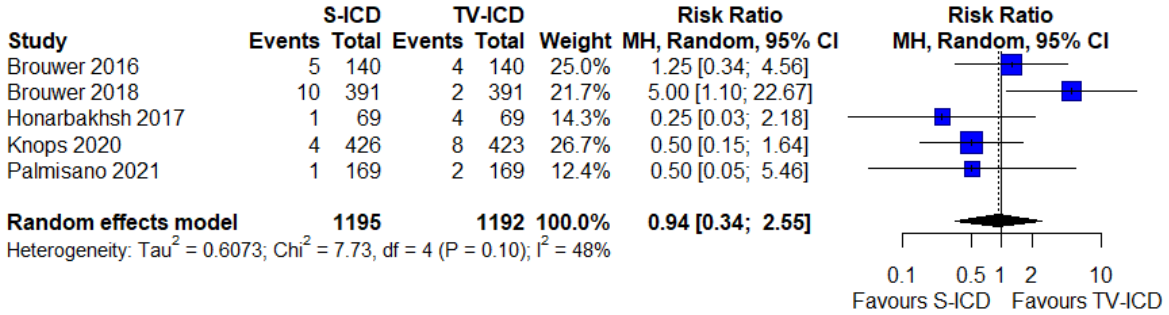


Figure S5. Forest plot of inappropriate shocks.

Study	S-ICD Events	S-ICD Total	TV-ICD Events	TV-ICD Total	Weight	Risk Ratio MH, Random, 95% CI
Brouwer 2016	20	140	22	140	30.2%	0.91 [0.52; 1.59]
Brouwer 2018	3	391	2	391	3.0%	1.50 [0.25; 8.93]
Honarbakhsh 2017	3	69	6	69	5.2%	0.50 [0.13; 1.92]
Knops 2020	41	426	29	423	45.3%	1.40 [0.89; 2.22]
Palmisano 2021	11	169	14	169	16.3%	0.79 [0.37; 1.68]
Random effects model		**1195**		**1192**	**100.0%**	**1.06 [0.78; 1.45]**

Heterogeneity: $Tau^2 = 0$; $Chi^2 = 3.69$, df = 4 (P = 0.45); $I^2 = 0\%$

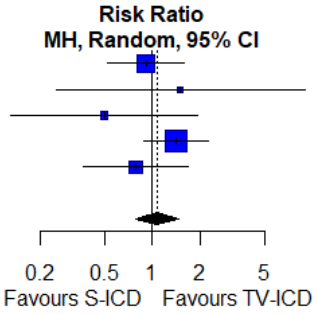


Figure S6. Forest plot of cardiac oversensing as a cause of inappropriate shocks.

Study	S-ICD Events	S-ICD Total	TV-ICD Events	TV-ICD Total	Weight	Risk Ratio MH, Random, 95% CI
Brouwer 2016	17	140	1	140	25.9%	17.00 [2.29; 126.01]
Brouwer 2018	2	391	0	391	11.3%	5.00 [0.24; 103.81]
Knops 2020	24	426	2	423	50.5%	11.92 [2.83; 50.10]
Palmisano 2021	4	169	0	169	12.3%	9.00 [0.49; 165.87]
Random effects model		**1126**		**1123**	**100.0%**	**11.44 [4.12; 31.74]**

Heterogeneity: $Tau^2 = 0$; $Chi^2 = 0.47$, df = 3 (P = 0.92); $I^2 = 0\%$

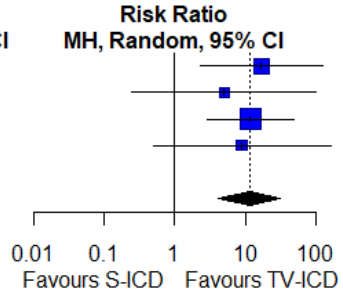


Figure S7. Forest plot of supraventricular tachycardia or atrial fibrillation as a cause of inappropriate shocks.

Study	S-ICD Events	S-ICD Total	TV-ICD Events	TV-ICD Total	Weight	Risk Ratio MH, Random, 95% CI
Brouwer 2016	3	140	21	140	22.3%	0.14 [0.04; 0.47]
Brouwer 2018	0	391	1	391	3.1%	0.33 [0.01; 8.16]
Honarbakhsh 2017	0	69	6	69	3.9%	0.08 [0.00; 1.34]
Knops 2020	11	426	27	423	63.0%	0.40 [0.20; 0.80]
Palmisano 2021	1	169	9	169	7.6%	0.11 [0.01; 0.87]
Random effects model		**1195**		**1192**	**100.0%**	**0.27 [0.15; 0.48]**

Heterogeneity: $Tau^2 = 0.0102$; $Chi^2 = 4.07$, df = 4 (P = 0.40); $I^2 = 2\%$

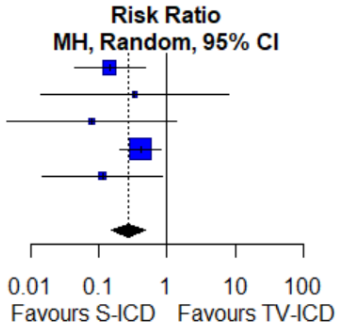


Figure S8. Forest plot of appropriate shock therapy.

